ATLAS INSTITUTIONAL CRYPTO SCREENER

SYSTEM ARCHITECTURE & QUANTITATIVE STRATEGY SPECIFICATION (V2.4)

Status: Production Institutional Release
Venue: Binance USD-M & Spot Universes
Execution: Multi-Timeframe Algorithmic Radar

1. EXECUTIVE SUMMARY & THE QUANTITATIVE EDGE

The Atlas Institutional Crypto Screener is a multi-timeframe quantitative screening engine designed to systematically identify high-expectancy trade opportunities across Binance USD-M perpetual contracts. Rather than chasing post-expansion volatility (+20% to +50% pumps) where retail traders routinely become exit liquidity, Atlas identifies:

- **Long Opportunities:** Pre-breakout volatility coils, Wyckoff Phase C springs, and quiet volume-profile absorption *before* trend markup.
- **Short Opportunities:** Post-vertical run-up blow-off tops, Wyckoff Phase C Upthrusts After Distribution (UTAD), and multi-timeframe momentum exhaustion in the upper 55% to 100% of the 30-day range.

2. CLEAN HEXAGONAL SYSTEM ARCHITECTURE

The platform is structured using strict Hexagonal (Ports & Adapters) architectural separation to isolate data feeds, quantitative mathematical modeling, and execution logic:

1. Domain Layer

Pure business models: `CryptoOpportunity`, `Direction`, `SetupLabel`, `Candle`, `IndicatorSnapshot`, and `ChecklistVerification`. Zero external dependencies.

2. Application Core

`CryptoOpportunityScreener` and `AlgorithmicTradingSystem` orchestrate multi-factor scoring, Wyckoff classification, and execution bracket construction.

3. Adapters & API

Binance USD-M client, CCXT bridge, DeFiLlama stablecoin telemetry, Alternative.me Fear & Greed API, and high-performance FastAPI asynchronous REST/WSS endpoints.

3. UNIVERSE SELECTION & LIQUIDITY FILTERING PIPELINE

From an unfiltered raw universe of 740+ instruments on Binance, the screening pipeline applies rigorous multi-stage hygiene gates before computing quantitative factors:

Pipeline Stage	Filtering Mechanism	Institutional Rationale
1. Asset Class Gate	Only USDT and USDC perpetual futures; trading enabled.	Ensures deep programmatic derivative liquidity and continuous two-sided order books.
2. Toxic Asset Blacklist	Excludes gold pegs (`PAXG`), synthetic commodities (`XAU` , `COPPER`), dead coins (`WLFI`), and stablecoin pairs.	Eliminates non-crypto volatility profiles and illiquid abandoned project anomalies.
3. Multiplier & Symbol Hygiene	Strict ASCII validation; blocks 1000x meme multiplier contracts (`1000SHIB` , `1000LUNC`).	Prevents tick-size math distortion and decimal rounding slippage during execution bracket sizing.
4. Liquidity & Spread Floor	Minimum 24h quote volume of **$500,000**; maximum bid-ask spread capped at **50 bps**.	Eliminates illiquid wash-trading pairs vulnerable to catastrophic stop-loss slippage.
5. Top ADV Stratification	Sorts liquid universe descending by 24h volume and selects the **top 160 liquid USD-M instruments**.	Focuses processing power and derivatives queries on assets with deep institutional participation ($20M to $1.2B ADV).

4. MULTI-TIMEFRAME TELEMETRY PIPELINE

For every selected instrument, Atlas concurrently ingests **Daily, 4-Hour, and 1-Hour candle series**, evaluating:

> **Ingested Signals:** 1H Indicator Snapshot (EMA9/20/50/200, ATR14, MACD, ADX, DMI, Normalized Momentum) • 4H Multi-Week Base Structure (EMA20/50, Keltner Channels, Bollinger Bands, Choppiness Index) • Volume Profile (50-bin VPVR POC, VAH, VAL) • 7-Day / 1-Hour Open Interest Delta • Funding Rate APR • Taker Buy/Sell Aggression.

QUANTITATIVE LONG SCREENING STRATEGY

PRE-BREAKOUT VOLATILITY SQUEEZE & WYCKOFF PHASE C ACCUMULATION

Direction: LONG Setups
Horizon: Multi-Day Swing (1–5 Days)
Min Conviction: 70.0 / 100.0 Floor

1. THE PRE-BREAKOUT PHILOSOPHY

Traditional retail screeners alert on tokens *after* they appear on "Top Gainers" (+20% to +40% daily pumps). By that point, smart money is actively distributing inventory into breakout chasers. The Atlas Long engine operates on the opposite principle: identify **volatility compression**, **supply exhaustion**, and **order-flow absorption** *before* the volatility expansion occurs.

2. THE 6 MANDATORY QUANTITATIVE LONG GATES

Gate Name	Mathematical Formulation	Institutional Function
Gate 1: Minervini Trend Baseline	`Close >= 4H_EMA50 * 0.985` `4H_EMA20 >= 4H_EMA50 * 0.975`	Mandates Stage 1 accumulation or Stage 2 markup. Eliminates Stage 4 downtrends and falling knives.

Gate 2: Anti-Chasing 2.0	`Excursion_72h <= 36%`
	`Ret_24h in [-8%, +14%]`
	`RSI_1H <= 76 & RSI_4H <= 70`
		Disqualifies parabolic blow-off pumps and post-pump exhaustion wicks. Ensures price is tightly coiled.

Gate 3: Base Duration & Ceiling Proximity	`Range_Height_30d <= 65%` `Dist_to_Ceiling in [-3%, +35%]`	Verifies multi-week consolidation base. Price must be resting beneath resistance, ready for expansion.
Gate 4: Value Area Acceptance	`Close >= 30D_POC * 0.975` `Value_Area_Pos >= 25%`	Confirms price is accepted above the 30-day Point of Control. Rejects tokens trading beneath heavy overhead volume.
Gate 5: Volatility Compression	`BB_Inside_KC == True` or `Bandwidth < 28% (Ideal < 10%)`	Identifies John Carter Squeeze geometry: Bollinger Bands contracting completely inside Keltner Channels.
Gate 6: S/R Breakout Resistance Gate	`if Dist_to_Ceiling <= 4.5%:` `  OI_7d >= -1% & RVOL >= 1.15x`	**Critical Bull-Trap Filter:** If price tests ceiling resistance while OI is bleeding and volume is flat, it is rejected immediately. Breakouts require expanding OI.

3. LONG SETUP ARCHETYPES

`COIL_SPRING_LONG` **Wyckoff Phase C**

Price conducts a terminal shakeout/spring test below support, rapidly reclaims the 4H EMA50 / POC shelf, with down-volume drying up (VCR ≤ 0.90x).

`BREAKOUT_LONG` **Volume Expansion**

Price breaks above local swing pivot / VAH ceiling with relative volume expansion (≥ 1.25x), positive normalized momentum, and expanding open interest.

`MOMENTUM_EMERGING_LONG` **Residual Alpha**

Early phase regime shift with strong positive BTC-residual alpha (≥ +3% over 24h), bullish MACD histogram expansion, and aggressive taker buying (>1.15x).

`OI_ACCUMULATION_LONG` **Smart Money Absorption**

Continuous institutional accumulation characterized by massive 7-day open interest expansion (+10% to +60%) while price remains compressed within a tight base.

4. EXECUTION BRACKET CONSTRUCTION (ASYMMETRIC R:R > 2.5:1)

Every long candidate is stamped with a deterministic 5-level execution bracket:

- **Retest Entry (Limit):** Anchored at the 4H EMA50 / POC support shelf (`Primary_Shelf + 0.15 * ATR`).
- **Breakout Trigger (Stop-Buy):** Placed immediately above local compression pivot resistance (`Resistance + 0.05 * ATR`).
- **Invalidation Stop Loss:** Placed structurally below the recent 4H swing low shelf (`Support - 0.35 * ATR`).
- **Multi-Tier Targets: Target 1:** 30d High / VAH ceiling • **Target 2:** Fibonacci 1.618 expansion • **Target 3:** 2.618 runner.

QUANTITATIVE SHORT SCREENING STRATEGY

VERTICAL RUN-UP BLOW-OFFS, WYCKOFF UTAD & UPPER-RANGE DISTRIBUTION

Direction: SHORT Setups
Horizon: Multi-Day Swing (1–5 Days)
Min Conviction: 70.0 / 100.0 Floor

1. THE SHORT ENGINE PHILOSOPHY: NEVER SHORT SUPPORT FLOORS

A common flaw in algorithmic screeners is shorting assets that have already suffered severe multi-week downtrends and are grinding into support floors (e.g. TRB, AXS, HBAR at range lows). These setups offer poor risk/reward and suffer vicious short-squeeze spikes. The Atlas Short engine enforces that **shorts are sold HIGH**—specifically targeting overextended assets in the **upper 55% to 100% of their 30-day range following a confirmed vertical run-up** where late retail buyers are trapped.

2. THE 5 MANDATORY QUANTITATIVE SHORT GATES

Gate Name	Mathematical Formulation	Institutional Function
Gate 1: Mandatory Macro Run-Up	`Range_Height >= 16%` or `Ret_7d >= 10%` or `Ret_14d >= 14%` or `(Close - EMA50_4H) >= +8%`	**Distribution requires prior markup.** Eliminates flat, lifeless, range-bound tokens that never experienced speculative vertical expansion.
Gate 2: Strict Upper-Range Positioning	`Pos_in_Range = (Close - Low_30d) / (High_30d - Low_30d)` **Mandate:** `Pos_in_Range >= 0.55`	Guarantees shorts are initiated strictly in the upper half of the range near distribution ceilings. Rejects bottom-chopping tokens.
Gate 3: Anti-Chasing & Oversold Filter	`Ret_24h >= -18% & Ret_7d >= -35%` `RSI_1H >= 28 & RSI_4H >= 38`	Prevents shorting into deeply exhausted bottoms or selling into capitulation wicks where mean-reversion bounces occur.
Gate 4: Multi-Timeframe Bearish Divergence	`RSI_4H < Prev_RSI_4H` on Higher High 4H MACD Histogram Bearish Divergence	Detects institutional exhaustion: price prints a nominal higher high, but momentum indicators decisively print lower peaks.
Gate 5: Trapped Longs & OI Crowding	`Funding_Rate > 0` (Longs paying) Large positive 7d OI (+15% to +200%)	Identifies overcrowded retail longs paying funding fees into smart-money distribution wicks, providing fuel for cascading liquidations.

3. SHORT SETUP ARCHETYPES

`UTAD_DIVERGENCE_SHORT` **Phase C**

Wyckoff Upthrust After Distribution: Price sweeps the 30-day high with buying climax wicks, followed by confirmed 4H/Daily bearish divergence and immediate rejection back inside the range.

`DISTRIBUTION_SHORT` **Macro Rollover**

Parabolic Markup Breakdown: Multi-week runners (>75% to 240% expansion) that coil below the 4H EMA50 and 30-day POC, displaying heavy seller volume dominance and structural momentum decay.

`EXHAUSTION_REVERSAL_SHORT` **Climax**

Multi-Timeframe Overbought Blow-Off: Assets where Weekly and Daily RSI exceed 75–84 with massive open interest surges (+50% to +216% 7d OI), ready for an immediate mean-reversion cascade.

4. SHORT EXECUTION HIERARCHY & INVALIDATION RULES

Short execution mandates strict geometric discipline to eliminate squeeze exposure:

```
EXECUTION HIERARCHY: Invalidation Stop > Retest Entry (EMA50/POC) >= Current Price > Breakdown Trigger > Downside Targets
```

- **Retest Entry (Sell-Limit):** Set at the 4H EMA50 / POC retest shelf (`POC_Price - 0.15 * ATR`) for maximum R:R (>5.5:1).
- **Breakdown Trigger (Stop-Market):** Set below the local consolidation floor (`Local_Low - 0.05 * ATR`) for momentum capture.
- **Invalidation Stop:** Placed structurally above the local distribution swing high / VAH ceiling (`Resistance + 0.35 * ATR`).

SCORING ENGINE & MATHEMATICAL EXPECTANCY

MULTI-FACTOR REGRESSION, COMPOSITE CONVICTION & EV OPTIMIZATION

Model: Non-Linear Factor Weighting
Metric: Expected Value in R (EV_R)
Threshold: 5-Point Verification Matrix

1. MULTI-FACTOR REGRESSION FORMULA

Every candidate passing the hard structural gates is evaluated through a non-linear composite scoring engine (0.0 to 100.0) across five orthogonal quantitative dimensions:

```
SCORE = 0.25 * Trend_Score + 0.25 * Momentum_Score + 0.25 * Derivatives_Score + 0.15 * Alpha_Score + 0.10 * Volume_Score
```

Factor Dimension	Weight	Underlying Quantitative Metric Inputs
Trend Architecture	25%	4H EMA20 vs EMA50 alignment, ADX trend strength (≥25), DMI directionality, Minervini stage positioning.
Momentum & Squeeze	25%	Bollinger Bandwidth (<10%), BB nested inside KC, MACD histogram velocity, normalized price momentum.
Derivatives Positioning	25%	7-day & 1-hour Open Interest velocity, contrarian funding rate arbitrage, taker buy/sell ratio, order book depth imbalance.
BTC-Residual Alpha	15%	Residual alpha vs Bitcoin over 6h, 24h, and 7d horizons; beta adjustment to isolate idiosyncratic token momentum.
Volume Microstructure	10%	Relative volume surge (RVOL ≥ 1.25x), volume contraction ratio (down-bar dry-up), 24h quote volume depth.

2. COMPOSITE CONVICTION RANKING

To prevent ranking bias between high-win-rate conservative setups and high-R:R aggressive setups, Atlas ranks candidates by **Composite Conviction**:

```
CONVICTION = 0.40 * Estimated_Win_Rate_Pct + 0.60 * Quantitative_Score
```

The system enforces a **70.0 minimum score threshold**. Low-conviction noise (scores 40–60) is filtered out before reaching the operator dashboard.

3. THE MATHEMATICAL EXPECTANCY MODEL (EV_R)

Trade viability is judged not by win rate alone, but by mathematical expected value per unit of risk (R). The platform calculates expected return using:

```
EV_R = [ P_win * (0.60 * R_T1 + 0.40 * R_T2) ] - [ (1 - P_win) * 1.0 ]
```

Where R_{T1} and R_{T2} represent the risk-adjusted payoff at Target 1 and Target 2, assuming a 1.0R loss on invalidation. Trades must achieve $EV_R \ge +1.50R$ to earn a Grade A recommendation.

4. THE 5-POINT INSTITUTIONAL CHECKLIST

Every candidate must undergo a deterministic 5-point verification audit displayed as badges on the trader dashboard:

1. VALUE ACCEPTANCE / BREAKDOWN
Verified trading above 30d POC (Long) or rejected below 30d POC / 4H EMA50 (Short).

2. VOLATILITY COMPRESSION / UTAD
Verified BB inside KC squeeze (Long) or confirmed 4H/Daily bearish divergence (Short).

3. MOMENTUM & TREND ALIGNMENT
4H EMA20/50 alignment and MACD histogram acceleration in trade direction.

4. VOLUME CONTRACTION / EXHAUSTION
Volume drying up on pullbacks (VCR ≤ 1.0x) or volume surge on breakout trigger.

5. SMART MONEY ORDER FLOW & DERIVATIVES POSITIONING
Non-negative 7d OI on long breakouts; trapped overleveraged retail longs (+20% to +200% OI, positive funding) on short setups.

MACRO INTELLIGENCE & OPERATIONAL TOPOLOGY

MARKET OVEREXTENSION ARCHITECTURE, REAL-TIME TELEMETRY & SYSTEM RESILIENCE

Service: Macro Liquidity & Radar Port
Cadence: 8-Hour UTC Funding Aligned Cron
Stack: Python 3.13 / FastAPI / Next.js 16

1. MACRO OVEREXTENSION ARCHITECTURE (MOEA)

The platform incorporates a Macro Market Intelligence engine to determine overall market regimes and compute the probability of a market-wide sell-off or cool-off. The **Composite Overextension Index (COI: 0–100)** unifies four distinct market pillars:

```
COI = 0.35 * Benchmark_Excursion + 0.30 * Leverage_Crowding + 0.20 * Market_Breadth + 0.15 * Stablecoin_Liquidity
```

1. Benchmark Excursion (35%)

BTC & ETH distance from Daily EMA50 and Daily Bollinger Upper Band, plus 14-day RSI excursion.

2. Leverage & Funding Crowding (30%)

Annualized perpetual funding rates across the top 20 perpetuals. Rates > 25% APR indicate extreme long vulnerability.

3. Market Breadth & Euphoria (20%)

Percentage of altcoins trading above their 20-day moving average combined with Alternative.me Fear & Greed index.

4. Stablecoin Fuel vs Leverage (15%)

7-day net issuance flow of USDT/USDC via DeFiLlama. Divergence between price gains and stablecoin outflows signals synthetic leverage.

2. AUTOMATED 8-HOUR CRON SERVICE ALIGNED WITH FUNDING BOUNDARIES

The screener background daemon executes every 8 hours, synchronized precisely with global Binance USD-M funding settlement boundaries:

- **Settlement Boundaries: 00:00 UTC · 08:00 UTC · 16:00 UTC** (+30 second settlement grace).
- **Operational Rationale:** Institutional funding fee payments reset derivatives positioning at these exact intervals, creating the cleanest multi-day directional setups.
- **Instant Client Caching:** Results are cached to `data/screener_latest.json` and served in <10ms to trader dashboards.

3. PRODUCTION INFRASTRUCTURE & RESILIENCE MATRIX

Infrastructure Layer	Technology Implementation	Fault Isolation Guarantee
Backend Core	Python 3.13 / FastAPI asynchronous runtime with Uvicorn.	Non-blocking async worker pool with rate-limit budgeting (semaphore = 10).
Frontend Portal	Next.js 16 (Turbopack) / React 19 / Canvas Charting.	Zero-lag HTML5 Canvas chart engine; virtualized candidate grids with zero layout shift.
Real-Time Ticker	Combined Binance WebSocket Multi-Ticker Stream.	Instant sub-second price updates with automatic REST fallback polling upon connection loss.
External APIs	Async fallback wrappers for DeFiLlama & CoinGecko.	Graceful degradation to cached baseline metrics if external third-party endpoints timeout.